

Association for Information Systems

AIS Electronic Library (AISeL)

AMCIS 2024 Proceedings

Americas Conference on Information Systems
(AMCIS)

August 2024

COF-MED-LSTM: Integrating Multimodal Events and LSTM Networks for Enhanced Crude Oil Futures Forecasting

Jinshui Huang

School of Management Science and Engineering, huangjinshui341000@163.com

Follow this and additional works at: <https://aisel.aisnet.org/amcis2024>

Recommended Citation

Huang, Jinshui, "COF-MED-LSTM: Integrating Multimodal Events and LSTM Networks for Enhanced Crude Oil Futures Forecasting" (2024). *AMCIS 2024 Proceedings*. 13.

https://aisel.aisnet.org/amcis2024/sig_dite/sig_dite/13

This material is brought to you by the Americas Conference on Information Systems (AMCIS) at AIS Electronic Library (AISeL). It has been accepted for inclusion in AMCIS 2024 Proceedings by an authorized administrator of AIS Electronic Library (AISeL). For more information, please contact elibrary@aisnet.org.

COF-MED-LSTM: Integrating Multimodal Events and LSTM Networks for Enhanced Crude Oil Futures Forecasting

Completed Research Full Paper

Jinshui Huang

Southwestern University of Finance and Economics

huangjinshui341000@163.com

Abstract

This paper uses the COF-MED-LSTM model, an advanced approach integrating multi-modal event-driven data for predicting crude oil futures prices. By leveraging distributed multi-thread crawlers for extensive data collection and employing the transformer's multi-head attention mechanism for feature extraction, this research overcomes the challenges of sampling heterogeneity between structured historical data and unstructured news information. The event-driven memory mechanism addresses data dimensionality and sampling issues, enhancing prediction accuracy. Empirical analysis across various periods demonstrates its superiority over traditional and advanced algorithms. This study contributes to crude oil futures theories and suggests further exploration into the indirect effects of news on related futures, highlighting the potential for model optimization and a deeper market understanding.

Keywords

Crude Oil Futures Prediction, Event-Driven LSTM, Multimodal Data Integration.

Introduction

In the complex landscape of global economics, crude oil stands as a linchpin, driving the development of nations and dictating the tempo of industrial progress. Within the vibrant ecosystem of the futures market, crude oil emerges as a pivotal element, serving dual critical functions: mitigating risk and facilitating price discovery. As the most traded commodity futures, crude oil futures represent a complex, high-stakes arena, underscored by the tremendous \$14.44 trillion trading volume on the Brent crude futures alone in 2023, as reported by the American Futures Association. This substantial market activity, coupled with the significant trading volume of Shanghai crude oil futures — surpassing 40 trillion yuan in 2023 — underscores the immense economic impact and the crucial role of futures price forecasting. As one of the core elements of futures trading, the crude oil futures price reflects the operation of the entire futures market (Richard & Sundaresan 1981; Li et al., 2020). Therefore, the research on futures pricing is of great significance, and has always been a classic proposition.

Historically, the prediction of crude oil futures prices has leaned heavily on historical trading price (Casassus et al., 2013; Gorton et al., 2013; Ma et al., 2019), trading volume (Roon et al., 2000), and open interest positions (Roon et al., 2000; Dewally et al., 2013; Ma et al., 2019). However, the increasing complexity of the market and the growing influence of global events, such as geopolitical shifts and economic upheavals, have highlighted the limitations of relying solely on historical data. The unprecedented negative pricing of WTI crude oil futures in April 2020, driven by the COVID-19 pandemic, is a stark testament to the market's sensitivity to real-time news and events, prompting a reevaluation of predictive methodologies to incorporate unstructured data sources like news information.

In addition, when integrating historical trading information and news text information and exploring the volatility of the crude oil futures market, there is still a problem that is difficult to solve, namely, the problem of uneven data sampling from different data dimensions and different information sources. Especially for the futures market, the historical trading data is uniformly sampled in time, with a fixed

time interval (such as every day, every minute, etc.). For news text information, the time of publication is uncertain, and the sampling has discontinuous time interval (Li et al., 2020). A more intuitive understanding is that the news of crude oil futures would not be released on every day with trading data. For this problem, it was mainly solved by using partial data, that is, using historical data matching the news release time to predict future prices, and directly discarding other transaction data without corresponding news. For example, Sadik uses historical trading data of the news date of the day to analyze the futures market in combination with the news (Sadik et al., 2020). However, this method of using partial data transaction data is easy to cause the loss of temporal characteristics of historical transaction data. Therefore, how to overcome the adverse effects caused by sampling imbalance when fusing data of different dimensions is very important for capturing the volatility of crude oil futures market.

This paper uses the Crude Oil Futures Event-driven Long Short-Term Memory (COF-MED-LSTM) Model, a pioneering approach that seeks to transcend traditional forecasting limitations by integrating multimodal data sources to achieve the following three goals:

- Using distributed multi-thread crawlers to crawl the massive crude oil futures news text from the network media sites, and form the crude oil futures market financial text database;
- Introducing news into the study of volatility of crude oil futures, and integrate emotional quantification method with historical trading data to become a more valuable and representative feature;
- The memory control of Event-driven LSTM model is carried out to solve the problem of uneven sampling of data in different dimensions (first proposed by Li et al. (2020)). This strategic mechanism embedded in our model could handle the inherent disparities in temporal distribution between structured historical data and the stochastic nature of news information, thereby preserving integrity and value of multimodal inputs in forecasting futures prices.

Related Work

In comprehensive review of 95 economics papers and 32 computer science papers, this paper delves into the scholarly discourse from two critical vantage points. Initially, the paper explore the evolution of feature extraction techniques, transitioning from structured data's numerical representations to the vectorization of text within high-dimensional, unstructured datasets. Subsequently, we navigate the progression in model architecture—from economic models to AI methodologies. This bifurcated review not only encapsulates the breadth of methodological innovation but also underscores the interdisciplinary approach pivotal in deciphering the complexities inherent in future market predictions.

Feature Selection

In financial forecasting, the majority of preceding studies have primarily utilized time series analysis models, focusing on numerical indicators such as futures prices, trading volumes, and open positions as foundational inputs. These models consider the influence of preceding input data sequences on subsequent output results. For instance, Baruník and Malinská (2016) validated the practicality of commodity future pricing theories through an analysis of position data from 3 energy futures markets.

Traditional financial time series prediction models such as AR, MA, and ARMA have showcased obvious linear characteristics. However, they fall short in capturing the latent nonlinear components within the time series data. This gap has spurred the development of nonlinear hybrid machine learning models that incorporate time series attributes. Wang et al. (2019) introduced an innovative DPFWR neural network algorithm, integrating a dual parallel forward neural network with wavelet analysis, including a random time-effect function, to weigh the impact of historical price information across different periods. Similarly, Creator and Lee (2019) developed a Multivariate Distance Nonlinear Causality test (MDNC) methodology to predict major energy financial time series, including oil futures, coal futures, and natural gas futures. Gu et al. (2019) modified the RNN structure utilizing LSTM to enhance the transmission of prior information to subsequent events, with empirical research validating the model's predictive capability using statistical tests on soybean futures market data from China (2010-2017).

In the digital age, the surging influence of media information has been recognized as a significant factor affecting investor behavior and, consequently, crude oil futures trends. The conventional models, characterized by low-dimensional numerical scales, are inadequate for capturing the evolving market.

Addressing this challenge, some studies have refined and augmented existing models with high-dimensional news information text vectors, paving new avenues for applying machine learning techniques to futures trading research. Sadik et al. (2020) introduced a method to incorporate macroeconomic news into crude oil futures price models, significantly enhancing prediction accuracy. Prusa et al. (2017) leveraged features extracted from the International Energy Agency's monthly reports to predict crude oil futures valuations, demonstrating that text features improve predictive performance.

Further advancing the field, Li et al. (2019) employed Convolutional Neural Networks (CNN) to unearth latent patterns in online media, proposing a feature grouping method based on the Latent Dirichlet Allocation (LDA) topic model to differentiate predictive effects of various topics. Their empirical research corroborated that integrating text and financial features yields more precise crude oil price forecasts.

This paper leverages distributed multi-threaded crawlers to obtain a vast array of crude oil futures news text from online media sources, integrating it with foundational data to construct a comprehensive feature database for crude oil futures market. By introducing news text into the volatility analysis of crude oil futures market and integrating it with historical trading data through a sentiment quantification method based on the transformer's multi-head attention mechanism, this study aims to enrich the feature set, making it more valuable and representative for predictive analysis of crude oil futures.

Model Method

To elucidate the complex relationship between market signals and futures prices, scholars have historically leaned on econometric models to dissect the dynamics between market factors and futures price volatility, thereby facilitating reasoned futures pricing. For instance, Hemler et al. (1991) constructed a general equilibrium model to decode stock index futures prices within a fluctuating interest rate and market environment, revealing a direct correlation between market volatility and stock index futures prices through an empirical analysis of New York Stock Exchange data. Similarly, Bessembinder et al. (1993) delved into the interrelations among trading volume, market volatility across eight futures markets, uncovering that unanticipated trading volume shocks significantly amplify market volatility.

Type	Literature	Model	Compared Model	Evaluating Indicator
Numeric Scalar Representation	Hemler and Longstaff (1991)	FTDNN	RW, AR(1), VAR(1)	MAE, MME, RMSE
	Kosowski et al. (2007)	LSTM-RNN	T-RNN	MAE, MAPE, RMSE
	Gorton et al. (2013)	SW-GRU	GRU, EEMD-SW-GRU	TIC, RMSE, MAPE, SMAPE
	Gu et al. (2019)	DPFWRNN	BP, DPF, LSTM, SARIMA	TIC, RMSE, MAPE, SMAPE, DS-CID
Text Vector Representation	Prusa et al. (2017)	LDA, CNN	LR, RF, SVR	MAE, RMSE
	Lin et al. (2018)	Random, ZeroR	NB, C4.5, C4.5D, MLP, SVM, KNN	F-Measure and G-Measure
	Creamer and Lee (2019)	MLE, Kalman Filtering	VAR	MAE, RMSE

Table 1. Representative Literature of Two Types

The entering of computer science scholars into this field has heralded the adoption of diverse machine learning models, enhancing the ability to capture the multifaceted influences on the futures market. These models excel in aggregating the effects of varied market determinants, thereby enabling more precise and effective futures price predictions. Noteworthy is the work of Baruník and Malinská (2016), who elucidated the term structure of crude oil prices using the dynamic Nelson-Siegel model and a neural network-based regression framework for oil price predictions. Wang et al. (2019) introduced the DPFWR neural network algorithm, showcasing its prediction prowess through rigorous error analysis and the

novel DS-CID error estimation method. Building on this, Wang et al. (2020b) proposed the SW-GRU model, a hybrid forecasting approach that leverages empirical mode decomposition, outperforming existing models. Further, Wang et al. (2020a) developed a multi-hybrid predictive model that combines an array of advanced techniques, significantly enhancing data information capture. However, the complexity of feature engineering in traditional machine learning poses challenges, necessitating in-depth data exploration and dimensionality reduction before selecting optimal features for model training, a process hampered by ultra-high dimensional datasets.

The linkage between news information and crude oil futures market volatility encapsulates deep, nonlinear correlation patterns. Advancements in information technology have propelled scholars towards deep learning models to dissect these complex relationships. Deep neural networks, surpassing their "shallow" counterparts, offer profound insights into high-dimensional interrelations of market information on futures market volatility. Ding et al. (2006) pioneered a deep learning approach for event-driven futures market forecasting, employing neural tensor networks for event extraction from news texts and deep convolutional neural networks to gauge the event's impact. Lin et al. (2018) demonstrated deep learning model's superiority, achieving significant predictive improvements. Huang et al. (2017) utilized deep neural networks to assess Twitter sentiment and explored temporal futures data patterns via convolutional neural networks, highlighting the predictive enhancement provided by social media sentiment. Peng et al. (2018) applied word embedding and deep neural networks for futures news-based stock price prediction, markedly improving price and volatility forecasts without relying on historical data.

This paper introduces an event-driven memory mechanism to the LSTM crude oil futures forecasting model, addressing the challenges of uneven data sampling across dimensions and unveiling the nonlinear correlation patterns between news information and market volatility.

Model Framework

The dynamics of the crude oil futures market are influenced by a diverse array of informational inputs, spanning historical price data to media information. Historical data, characterized by continuous values sampled at uniform intervals, contrasts starkly with the sporadic nature of media information. This disparity in data uniformity can lead to a dilution of informational value. To mitigate this, we employ the transformer architecture for the articulation of market text information. Building on this, we use a Crude Oil Futures Multi-modal Event-Driven Long Short-Term Memory (COF-MED-LSTM) Model, designed to encapsulate the nonlinear relationship between market information and futures price trend. Figure 1 delineates the methodological schema underpinning our approach.

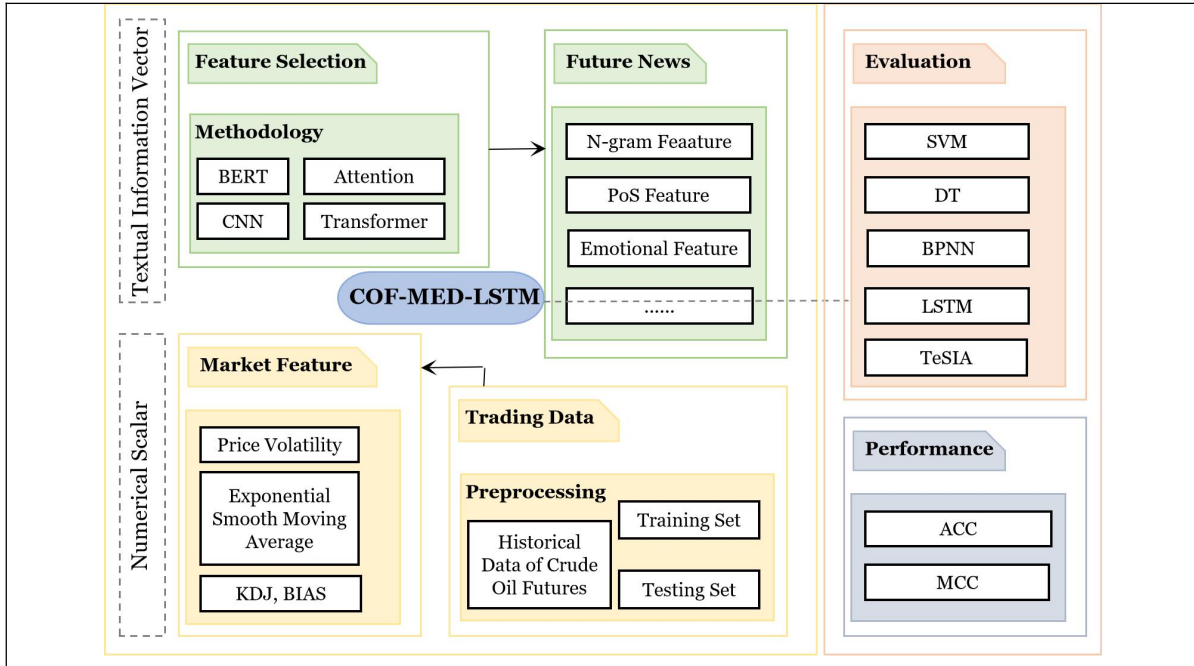


Figure 1. Overall Research Framework**Text Feature Extraction Technology**

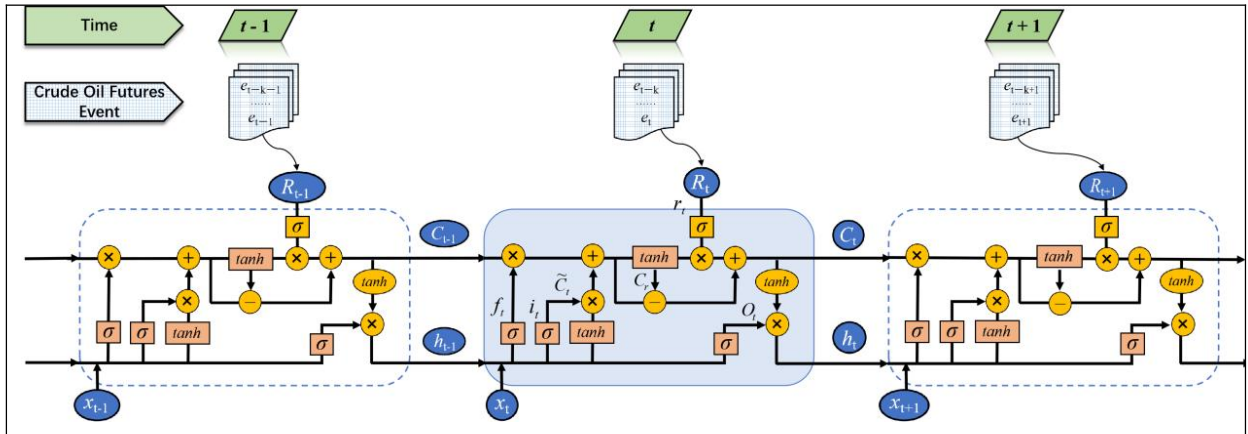
In acknowledgment of the vital role media information plays in the financial domain, we harness Natural Language Processing (NLP) technologies to distill relevant features into input vectors for deep learning models. Notably, the Transformer, BERT (Bidirectional Encoder Representations from Transformers), CNN (Convolutional Neural Networks), and attention mechanisms stand out as proficient methodologies for text representation. For the purposes of this study, we select the Transformer model to articulate media text information, particularly financial news related to crude oil futures, which is instrumental in conveying both macroeconomic fundamentals and analysts' sentiments.

Originating from Vaswani et al. (2017), the Transformer model signifies a departure from traditional encoder-decoder frameworks that relied on CNNs or RNNs. Instead, it exclusively employs attention mechanisms, eschewing CNNs and RNNs. This design choice positions the Transformer as a purely attention-based encoder-decoder model. The incorporation of self-attention within the Transformer introduces a novel architecture comprising layers of self-attention and fully connected layers. The model's essence is encapsulated in its encoder and decoder components. Each code unit is structured around two primary modules: a feed forward neural network and an attention mechanism, with the decoder incorporating an additional cross-attention mechanism. The quintessence of Transformer model resides in its innovative use of the attention mechanism, heralding a paradigm in text feature extraction.

Crude Oil Futures Multi-modal Event-driven LSTM Model

The market data is inherently multi-modal, encompassing both the continuous metrics of historical information and the discrete variables inherent in news narratives. Addressing the challenge of forecasting crude oil futures prices within the context of irregularly timed news, this paper uses the Crude Oil Futures Multi-modal Event-Driven Long Short-Term Memory (COF-MED-LSTM) model. This model can capture complex pricing mechanisms of the futures market and employs a trigger strategy to augment the event-based information. Figure 2 furnishes a comprehensive visualization of the model's architecture.

This model is meticulously designed to navigate the complexities of integrating diverse data forms—seamlessly synthesizing structured historical trading figures with the unstructured, sporadic influx of media reports (Li et al., 2020). By leveraging this dual-data approach, the COF-MED-LSTM model aspires to yield more accurate predictions of crude oil futures trends, enhancing understanding of market.

**Figure 2. Event-LSTM Model Memory Unit and Application Diagram in Analyzing Crude Oil Futures Market Information**

As shown in Figure 2, market information of time t is expressed in X_t . Event information E_t is represented by vector $\langle e_0, e_1, \dots, e_t \rangle$, where e_t is the total number of news articles at time t . There are two kinds of information flows to record the knowledge learned in the network. Specifically, the memory unit C_t records the event-based knowledge learned from the previous stage, and the output H_t records the

knowledge learned from the market information before time t . The forgetting gate F_t is used to control how much information should be retained or forgotten from the unit memory C_{t-1} at time t . In other words, F_t allows to discard the useless information related to H_{t-1} and X_t in C_{t-1} . An input gate I_t is used to control how much current information is absorbed into event knowledge flow C . In this structure, the memory unit C_t stores the knowledge learned from the current market information X_t and the previous stage information H_{t-1} through the mapping function in the neural network. Therefore, $\tilde{C}_t$ can capture valuable rules and patterns hidden in the previous and current time periods.

$$\tilde{C}_t = \tanh(W_c * X_t + U_c * H_{t-1} + V_c * E_t + B_c) \quad (1)$$

$$F_t = \sigma(W_f * X_t + U_f * H_{t-1} + V_f * E_t + B_f) \quad (2)$$

$$I_t = \sigma(W_i * X_t + U_i * H_{t-1} + V_i * E_t + B_i) \quad (3)$$

$$\hat{C}_t = F_t \circ C_{t-1} + I_t \circ \tilde{C}_t \quad (4)$$

In the formula, $\{W_c, U_c, V_c\}$, $\{W_f, U_f, V_f\}$, $\{W_i, U_i, V_i\}$ is the parameter of memory unit, forgetting gate and input gate, and B_c, B_f, B_i is the deviation of each; $\circ$ stands for Hadamar Product; $*$ represents a convolution operator. In order to deal with events with unequal time intervals, E_t is used to control which type of market information should be used according to the occurrence of events at time t . Therefore, we obtain the event control factor r_t through the non-increasing mapping function, as follows:

$$r_t = \sigma(V_r * E_t + B_r) \quad (5)$$

If there is a strong correlation between market information and current events, r_t is often very large. In this case, more event-related information C_r will be retained in the memory unit. Specifically, the event-related memory C_r can be extracted through the $\tanh$ function, and the memory unit C_t at time t is determined by equation (7):

$$C_r = \tanh(\hat{C}_t) \quad (6)$$

$$C_t = \hat{C}_t + (C_r \circ r_t - C_r) \quad (7)$$

Finally, the memory C_t and the market information X_t together obtain the output O_t through the output gate H_t . Specifically,

$$O_t = \sigma(W_o * X_t + U_o * H_{t-1} + V_o * E_t + B_o) \quad (8)$$

$$H_t = O_t \circ \tanh(C_t) \quad (9)$$

This architecture is adept at managing multi-modal data characterized by heterogeneous sampling intervals, specifically accommodating datasets where certain patterns are sampled at uniform intervals and others at non-uniform intervals (Li et al., 2020). This capability ensures comprehensive data integration, allowing for analysis across varied data streams, thus enhancing the model's predictive accuracy and robustness in handling the multidimensional financial market data.

Experimental Evaluation

To ascertain the efficacy of the COF-MED-LSTM model, which leverages multi-source information, we embarked on a comprehensive experimental study utilizing real-world trading data from three main crude oil futures markets: China, Brent, and WTI.

Experimental Data

Our analysis employed trading data from these markets alongside focused topic crawler technology to aggregate relevant news within crude oil futures field. The experimental data is detailed as follows:

- **Basic Data:** This dataset encompasses trading index data from Chinese crude oil futures (March 26, 2018, to September 25, 2022), Brent crude oil futures (December 29, 1990, to September 25, 2022), and WTI crude oil futures (November 19, 2010, to September 25, 2022). We employed pre- and post-average processing for missing values, with direct deletion utilized for front-end discrepancies.

- **Media Data:** The influence of significant news releases on investor expectations can precipitate market volatility. We collected 31,422 news within the same time span, pertinent to the crude oil futures, from www.china5e.com, a leading media in energy and futures information in China. Due to space constraints, Table 2, which provides descriptive statistics of historical trading data for the three major futures, is referenced for further exploration at [here](#).

The analysis revealed notable fluctuations in trading volume and positions, in contrast to the relatively modest changes in stock prices (opening, closing, highest, and lowest prices), with these price ranges predominantly circling around 600. This pattern aligns with expectations, given the capped crude oil futures price at 590.6 and significant trading volumes reaching up to 779,723 stocks. Moreover, crude oil futures price tends to orbit the spot price of crude oil and its intrinsic value, resulting in lower volatility.

For the experiment, initial 3/4 of the dataset served as the training set for the model, with the remaining 1/4 designated for evaluation. Notably, the evaluation employed a rolling window approach, wherein the model, initially trained on the first 3/4 of the data, was incrementally updated with input data preceding each evaluation day (t-day), ensuring incorporation of the latest market information into training model.

Evaluation Settings

This paper employ the standard measure of accuracy (ACC) and the Matthews correlation coefficient (MCC) as primary metrics to assess the model's performance. ACC, a widely utilized index in stock classification, gauges the congruence between predicted trends and actual stock price movements.

However, ACC may exhibit biases in datasets with significant class imbalances. For instance, in a sample set of 100, where 95% are positive instances, a classifier predicting all instances as positive would achieve a 95% ACC, despite its inability to identify negative instances. To counterbalance potential distortions stemming from data skewness, the MCC metric is incorporated, offering a more balanced performance evaluation. Higher values of both ACC and MCC indicate superior model performance.

Model Parameters

The algorithm presented is an advancement over the conventional LSTM framework, operationalized through Keras and TensorFlow for optimal efficacy. Parameter tuning was conducted to refine model's performance, with preliminary findings indicating an optimal θ value of 0.73. Among various activation functions explored—including sigmoid, tanh, and exponential linear unit (ELU)—the rectified linear unit (ReLU) function emerged as the most effective. The Adam optimizer was selected for adaptive learning rate capabilities, which are contingent upon historical update patterns of model weights.

Moreover, a series of pre-experimental analyses were undertaken to ascertain the optimal temporal impact of media influences within experimental setup. These investigations revealed that media effects could persist for up to a week, beyond which their influence on ACC and MCC metrics begins to diminish.

Model Comparison

To gauge the efficacy of COF-MED-LSTM model, we conducted a comparative analysis with several benchmark algorithms, notably including Support Vector Machine (SVM), Decision Tree, Backpropagation Neural Network (BPNN), and Long Short-Term Memory (LSTM) models. The baseline models are characterized as follows:

- **SVM:** This model integrates basic and media information into a comprehensive vector, utilizing it as the input. SVM operates as a generalized linear classifier that employs supervised learning to enact binary data classification, with its decision boundary constituted by the maximal margin hyperplane derived from the learning samples.
- **DT:** Recognized for its efficacy in price forecasting, the Decision Tree method forms part of benchmarking suite. It employs a concatenated vector directly inputted into the model for predictive analysis. This method generates a classifier based on a set of samples, each characterized by attributes and a predefined category, facilitating accurate classification of new entities to a certain degree.
- **BPNN:** This approach obviates need for a predetermined mathematical equation mapping input-output relationship, relying instead on learning rules through training to approximate the expected output

value given an input. Its foundation is gradient descent method, leveraging gradient search technology to minimize network's error mean square deviation between actual and expected output values.

- **LSTM:** The LSTM model is acclaimed for its proficiency with time series data, capturing temporal correlations in stock data using a concatenated vector as input. As a variant of RNN, LSTM is engineered to address the long-term dependency challenges inherent in conventional RNN.
- **TeSIA:** This method employs tensors to model market information, elucidating the interrelations among different information modalities. By representing basic and media information as tensors rather than vectors, it capitalizes on the multi-linear functionality of tensors to delineate the linear relationship among vectors, scalars, and other tensors.
- **COF-MED-LSTM:** Addressing the unique challenges posed by multi-source data — particularly harmonization of data sampling across disparate dimensions and information sources—model adeptly navigates the heterogeneity of data sampling intervals. It thereby significantly preserves valuable information by accommodating diverse temporal patterns inherent in data sources.

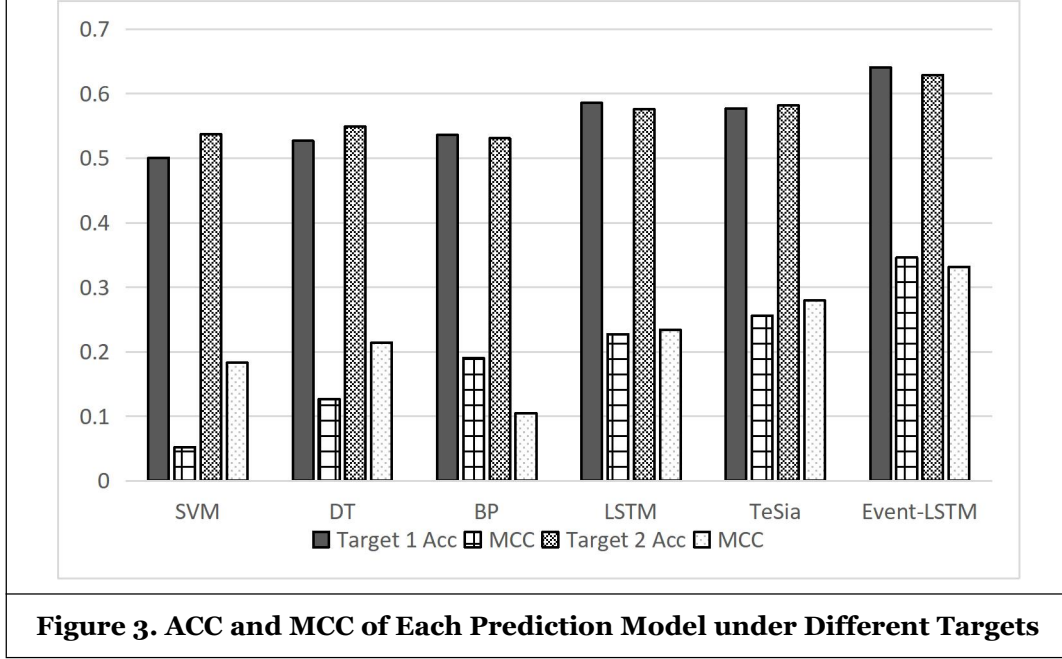
In this segment, we delineate the comparative efficacy of COF-MED-LSTM model against five classical algorithms: Support Vector Machine (SVM), Decision Tree (DT), Backpropagation Neural Network (BPNN), Long Short-Term Memory (LSTM), and a tensor-based method (TeSIA), across two distinct predictive targets. These experimental outcomes are encapsulated in Table 3.

Model	Target 1		Target 2	
	ACC	MCC	ACC	MCC
SVM	0.501	0.0524	0.537	0.1832
DT	0.527	0.1271	0.549	0.2143
BP	0.536	0.1903	0.531	0.1052
LSTM	0.586	0.2275	0.576	0.2343
TeSIA	0.577	0.2561	0.582	0.2799
COF-MED-LSTM	0.641	0.3462	0.629	0.3317
Table 3. Forecast Results of Futures Price Direction				

Effectiveness Analysis

The task of modeling crude oil futures movements, as influenced by media narratives, presents a quintessential multi-source information challenge. The crux of challenge lies in the heterogeneity of data sampling times—futures data being continuous and sampled at fixed intervals, juxtaposed against the stochastic nature of news information. This disparity not only risks the omission of critical information but can also skew the feature space, potentially compromising the LSTM model's ability to maintain long-term dependencies, especially when significant time lapses exist between related news articles.

Our analysis aims to evaluate the proficiency of the COF-MED-LSTM framework in surmounting these hurdles. As depicted in Figure 3, the COF-MED-LSTM model outperforms the traditional LSTM framework across all predictive targets, as evidenced by superior ACC and MCC metrics. This enhancement affirms the event-driven mechanism's capacity to unearth deeper insights and patterns within the crude oil futures market amidst random news occurrences. Prior attempts to address this issue often resorted to partial data utilization—retaining only the segments coinciding with news events for analysis. Such approaches, however, fall short of addressing the fundamental issue of sampling heterogeneity, leading to the potential loss of vital patterns and a diminished grasp of historical context. In contrast, our study uses an event-driven memory mechanism as an effective solution to the challenge of data source heterogeneity in the crude oil futures market, preserving valuable information and fostering a comprehensive understanding of market.



Conclusion

The crude oil futures market is significantly influenced by a plethora of information streams, spanning both traditional and behavioral finance perspectives. It is widely acknowledged that market information, encompassing both foundational data and news reports, plays a critical role in shaping the dynamics of crude oil spot and futures prices. Addressing this multifaceted information challenge necessitates a multi-source data approach, wherein each data type responds to specific attributes shared within its group. This study contemplates market information through two primary lenses: historical data and news text, confronting the critical challenge of sampling heterogeneity across these data modalities. The variability in data sampling—fixed intervals for basic data versus the randomness of news information—can obscure valuable insights and skew the analytical pattern.

To mitigate the discrepancies in data sampling across diverse information sources within the multimodal learning context of the crude oil futures market, this paper uses an event-driven memory mechanism. Empirical analysis conducted on international crude oil futures market data across various periods demonstrates the superiority of this approach over advanced algorithms like LSTM and Tesia, as well as traditional methodologies such as SVM and BPNN.

The novelty of this paper lies in its comprehensive approach: leveraging web crawler technology for extensive data collection of crude oil futures news, employing the transformer's multi-head attention mechanism for feature extraction and sentiment quantification from multi-source information, and applying advanced machine learning techniques to traditional financial market analysis. The findings of this study aim to contribute to the theoretical framework surrounding crude oil futures, encompassing futures storage theory, arbitrage limit theory, and term structure theory.

This paper, centered on the media's portrayal of crude oil futures trends, unveils the framework's applicability to a broad spectrum of multi-modal learning challenges. Furthermore, it addresses the gap in prior research that narrowly focused on direct impacts without considering the complex, indirect effects of news on related futures. This study prompts further exploration into the interconnected dynamics of futures markets and the potential for international market activities to influence futures variations, paving the way for model optimization and a deeper understanding of market mechanisms.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (NSFC), under Grants 6207070426, and Fundamental Research Funds for the Central Universities, under Grants JBK 2107015.

REFERENCES

- Baruník, J., Malinská, B., “Forecasting the term structure of crude oil futures prices with neural networks,” *Applied energy*, Vol. 164, pp. 366-379, 2016.
- Bessembinder, H., Seguin, P. J. “Price volatility, trading volume, and market depth: Evidence from futures markets,” *Journal of Financial and Quantitative Analysis*, pp. 21-39, 1993.
- Casassus, J., Liu, P., Tang, K., “Economic linkages, relative scarcity, and commodity futures returns,” *The Review of Financial Studies*, Vol. 26, No. 5, pp. 1324-1362, 2013.
- Creamer, G. G., Lee, C., “A multivariate distance nonlinear causality test based on partial distance correlation: a machine learning application to energy futures,” *Quantitative Finance*, Vol. 19, pp. 1531-1542, 2019.
- De Rooin, F. A., Nijman, T. E., Veld, C., “Hedging pressure effects in futures markets,” *The Journal of Finance*, Vol. 55, No. 3, pp. 1437-1456, 2000.
- Dewally, M., Ederington, L. H., Fernando, C. S., “Determinants of trader profits in commodity futures markets,” *The Review of Financial Studies*, Vol. 26, No. 10, pp. 2648-2683, 2013.
- Geng Ni, Brent crude oil futures price prediction based on wavelet transform and time series model [J], *China Business Journal*, Vol. 4, pp. 36-37, 2018
- Gorton, G. B., Hayashi, F., Rouwenhorst, K. G., “The fundamentals of commodity futures returns,” *Review of Finance*, Vol. 17, No. 1, pp. 35-105, 2013.
- Gu, Q., Lu, N., Liu, L., “A novel recurrent neural network algorithm with long short-term memory model for futures trading,” *Journal of Intelligent & Fuzzy Systems*, Vol. 37, No. 4, pp. 4477-4484, 2019.
- Hemler, M. L., Longstaff, F. A., “General equilibrium stock index futures prices: Theory and empirical evidence,” *Journal of Financial and Quantitative Analysis*, pp. 287-308, 1991.
- Hinton, G. E., Salakhutdinov, R. R., “Reducing the dimensionality of data with neural networks,” *science*, Vol. 313, No. 5786n 504-507. (2006).
- Li, Q., Tan, J., Wang, J., Chen, H. C., “A multimodal event-driven lstm model for stock prediction using online news,” *IEEE Transactions on Knowledge and Data Engineering*, forthcoming, 2020. DOI: 10.1109/TKDE.2020.2968894.
- Li, Q., Wang, T. J., Li, P., Liu, L., Gong, Q., Chen, Y., “The effect of news and public mood on stock movements,” *Information Sciences*, Vol. 278, pp. 826-840, 2014.
- Li, X., Shang, W., Wang, S., “Text-based crude oil price forecasting: A deep learning approach,” *International Journal of Forecasting*, Vol. 35, No. 4, pp. 1548-1560, 2019.
- Lin, C. B., Chou, R. K., Wang, G. H. K., “Investor sentiment and price discovery: evidence from the pricing dynamics between the futures and spot markets,” *Journal of Banking & Finance*, Vol. 90, pp. 17-31, 2018.
- Liu Song; Liu Hao; Yang Mengyuan, Impact effect of heterogeneous investor sentiment in international crude oil futures market under structural change [J], *International Business*, Vol 3, pp. 124-137, 2017.
- Ma Zhengwei; Zhang Jia; Cao Gaohang, Research on International Crude Oil Futures Price Fluctuation and Its Influencing Factors [J], *Price Theory and Practice*, Vol 4, pp. 87-91, 2019.
- MacKinlay, A. C., Ramaswamy, K., “Index-futures arbitrage and the behavior of stock index futures prices,” *The Review of Financial Studies*, Vol. 1, No. 2, pp. 137-158.
- Prusa, J. D., Sagul, R., Khoshgoftaar, T. M., Sterling, M., “Extracting Knowledge from Technical Reports for the Valuation of West Texas Intermediate Crude Oil Futures,” *IEEE International Conference on Information Reuse & Integration in 2017*.
- Richard, S. F., Sundaresan, M., “A continuous time equilibrium model of forward prices and futures prices in a multigood economy,” *Journal of Financial Economics*, Vol. 9, No. 4, pp. 347-371, 1981.
- Sadik, Z. A., Date, P. M., Mitra, G., “Forecasting crude oil futures prices using global macroeconomic news sentiment,” *IMA Journal of Management Mathematics*, Vol. 31, No.2, pp. 191-215, 2020.
- Wang, B., Wang, J., “Deep multi-hybrid forecasting system with random ewt extraction and variational learning rate algorithm for crude oil futures,” *Expert Systems with Applications*, Vol. 161, pp. 113686, 2020.
- Waswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... Polosukhin, I., “Attention is all you need,” In *Conference and Workshop on Neural Information Processing Systems(NIPS)*, 2017.
- Zhao, Y., Li, J., Yu, L., “A deep learning ensemble approach for crude oil price forecasting,” *Energy Economics*, Vol. 66, pp. 9-16, 2017.